O–L–1031a
April 13, 1967

SUPERSEDING
Int. Fed. Spec. O–L–001031 (Army–GL)
February 10, 1966

FEDERAL SPECIFICATION
LITHOGRAPHIC SOLVENT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type and one grade of lithographic solvent (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specifications:

P–D–680 —Dry Cleaning Solvent.
JJJ–C–86 —Castor Oil, Technical.
LLL–P–400—Pine Oil.
PPP–C–96 —Cans, Metal, 28 Gage and Lighter.

Federal Standards:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 141—Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

JAN–G–276—Gum Ester.

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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Federal Hazardous Substances Labeling Act.

(Application for copies should be addressed to the U.S. Department of Health, Education, and Welfare, Food and Drug Administration, Washington, D.C., 20203.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Pine oil. The pine oil shall be steam distilled, sulfate or synthetic type, and shall conform to the requirements of type I of LLL–P–400.

3.1.2 Castor oil. The castor oil shall be technical grade and shall conform to the requirements of grade 1 of JJJ–C–86.

3.1.3 Ester gum. The ester gum, a glycerol ester of rosin, shall conform to the requirements of type II, class d of JAN–G–276.

3.1.4 Drycleaning solvent. The drycleaning solvent shall conform to the requirements of type I of P–D–680.

3.2 Composition. The finished solvent shall conform to the requirements of table I, when tested as specified in 4.3.3.

TABLE I. Composition

Component	Percent by weight
Pine oil	9.4 ± 1.0
Castor oil	1.0 ± 0.1
Ester gum	2.0 ± 0.1
Drycleaning solvent	87.6 ± 2.0

3.3 Odor and color. The solvent shall have a pleasant pine odor and not darker than straw in color.

3.4 Flashpoint. The flashpoint of the solvent cleaner shall be not less than 100°F., when tested as specified in 4.3.3.

3.5 Labeling. Each unit container shall be durably and legibly marked in accordance with the Federal Hazardous Substances Labeling Act and shall be marked with the following information:

DIRECTIONS FOR USE

Remove dried and oxidized ink deposits and other stubborn soils from the press by rubbing the soiled parts with a pad of clean cloth dampened with solvent cleaner. Do not use this cleaner on rubber offset blankets or rollers because it may swell the rubber causing softening of the roller and delamination of the blanket.

3.5.1 Precaution. The following shall appear on all containers: Warning! Flammable. Keep away from heat, sparks, and open flame.

3.6 Workmanship. The finished product shall be clean, clear liquid, free from suspended and sedimentary matter.

4. QUALITY ASSURANCE PROVISION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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. 4.2 Inspection. Sampling for inspection shall be in accordance with MIL-STD-105, except where otherwise indicated.

4.2.1 Inspection of materials and components. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, or qualified in this specification or applicable purchase documents.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item and its preparation for delivery shall be examined for the defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.2. The lot size shall be expressed in unit containers of the specified capacity for the examination in 4.3.1.1, and in units of shipping containers for the examination in 4.3.1.2.

4.3.1.1 Examination of the end item for net contents. The sample unit for this examination shall be one filled unit container. The lot shall be unacceptable if the average net contents per container is less than specified or indicated for all sample units examined.

4.3.1.2 Examination of preparation for delivery. An examination shall be made to determine the packaging, packing, and markings comply with the requirements of this specification. The sample unit for this examination shall be one shipping container.

Examine	Defect
Packaging	Any leakage of contents. Any split, break, dent, puncture, or other defect.
Packing	Container not as specified. Any nonconforming, missing, damaged, or defective component.
Marking	Incorrect, illegible, omitted, of improper size, location, sequence, or method of application.

4.3.2 Inspection levels and AQLs for examination. The inspection levels for determining the sample size and the AQLs expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQLs
4.3.1.1	S-2	N.A.
4.3.1.2	S-1	2.5

4.3.3 Testing of the end item. The end item shall be tested for the characteristics specified in table II. The lot size shall be expressed in gallons of lithographic solvent. The sample for testing shall be a one pint composite made by selecting an equal amount from each of the containers, shown in the sampling below. The composite sample shall be thoroughly mixed, transferred to a clean, dry, glass container, sealed and used for tests. The lot shall be unacceptable if the composite fails to meet any of the test requirements specified.

Lot size (gallons)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

4.4 Test procedures.

4.4.1 Color and odor. After vigorously shaking the container of sample, fill a 100 mil graduate glass cylinder and immediately note odor; examine visually for color, clarity, and presence of suspended and sedimentary matter.

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TABLE II. Instructions for testing of the end item

Characteristic	Specification reference		Requirements - applicable to		Number determinations per unit	Results reported as		Inspect. level	AQL
	Require-ment	Test method	Individ. unit	Comp. samp.		Pass or fail	Numeri-cally to nearest		
Composition	3.2	[1]							
Odor	3.3	4.4.1		X	1	X			
Color	3.3	4.4.1		X	1	X			
Flashpoint	3.4	4291 [2]		X	Average of 2	X	°F.		

[1] Certificate of compliance will be furnished the contracting officer for composition requirements.
[2] Test performed in accordance to Fed. Test Method Std. No. 141.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. One-gallon of solvent shall be filled into a one-gallon capacity metal can conforming to type V, class 4, with a formed bridge-type handle securely affixed to the top, of PPP–C–96, except that 0.25-pound electrolytic tinplate, terneplate, and blackplate, shall not be used. The exterior coating shall conform to plan A.

5.1.2 Level C. Lithographic solvent shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Lithographic solvent, packaged as specified in 5.1, shall be packed in accordance with the level A packing requirements in the appendix of PPP–C–96.

5.2.2 Level B. Lithographic solvent, packaged as specified in 5.1, shall be packed in accordance with the level B packing requirements in the appendix of PPP–C–96.

5.2.3 Level C. Lithographic solvent, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Intended use. The lithographic solvent covered by this specification is intended to be used to remove dried and oxidized ink deposits and other stubborn soils from lithographic presses. It is not intended for use in cleaning rubber offset blankets.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).

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MILITARY INTEREST:

Custodians:

Army—GL
Navy—NP
Air Force—68

Review activities:

Army—GL, MD, MO, MU
Navy—NP
Air Force—68

Preparing activity:

Army—GL

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

CIVIL AGENCY INTEREST:

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